Homework — 1.2.1 Operating Systems — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] Abstraction = the OS hides the complexity of the underlying hardware from the user/programmer, providing a simpler interface (1). Platform = the OS provides a consistent environment/set of services on which applications can run, so software does not have to manage hardware directly (1).

2. [3] The CPU can only execute one process at a time on a single core (1); the scheduler rapidly switches between processes / allocates short turns of CPU time so each appears to run at once (time-slicing) (1). Two goals (1 mark for any two): maximise CPU usage / keep CPU busy; be fair so no process starves; give good response time to interactive jobs; maximise throughput. Max 3.

3. [3] FCFS runs processes in order of arrival until each completes (1); SJF runs the process with the shortest total runtime next, minimising average waiting time (1). Drawbacks (1 mark for both named): FCFS suffers the convoy effect — a long job at the front delays short jobs; SJF can starve long jobs and requires the runtime to be known in advance. Max 3.

4. [4] Virtual memory uses an area of secondary storage (the page/swap file) as if it were RAM (1); pages not currently needed are moved out of RAM to disk, freeing space so the larger program / more programs can run (1); when a page on disk is needed (page fault) it is swapped back into RAM (1). Disk thrashing occurs when so many pages are swapped between RAM and disk that the CPU spends most of its time paging rather than doing useful work, so the machine slows to a crawl (1).

5. [2] The BIOS is firmware on ROM that is the first program to run at start-up (1). Any two tasks (1 mark for two): performs the POST (power-on self-test) to check hardware is present/working; initialises hardware; reads low-level settings (clock, boot order); loads/bootstraps the OS into RAM. Max 2.

Part B (6 marks)

6. [2] PC holds the address of the next instruction to be fetched (1). MAR holds the address currently being accessed in memory (the address copied from the PC so the instruction can be fetched) (1).

7. [2] Multiple cores allow more than one instruction/thread to be executed truly in parallel, increasing throughput (1). It does not double performance because: not all tasks can be parallelised / some code is sequential; cores compete for shared resources (memory, cache, bus); overhead of coordinating/splitting work. Any one (1). Max 2.

8. [2] SSD advantage (1): faster access/read-write, no moving parts so more durable/silent, lower power. HDD advantage (1): cheaper per GB / larger capacity for the price. Max 2.

Total: 14 + 6 = 20